

IOT Based Smart Shopping Trolley

Prof. Pranavi Nikam
Department of Electronics
and Telecommunications
SIES Graduate School of
Technology
Navi Mumbai, India

Adhitya Kumaran
BE Electronics and
Telecommunications
SIES Graduate School of
Technology
Navi Mumbai, India

Harshita Shrivastav
BE Electronics and
Telecommunications
SIES Graduate School of
Technology
Navi Mumbai, India

Kavya Chandrasekaran
BE Electronics and
Telecommunications
SIES Graduate School of
Technology
Navi Mumbai, India

Krishnaveni Doki
BE Electronics and
Telecommunications
SIES Graduate School of
Technology
Navi Mumbai, India

Abstract—Currently, during holidays and weekends in urban areas, a huge number of shoppers can be seen crowding into shopping malls and supermarkets. When there are special offers and huge discounts, the number of people increases even more. In supermarkets these days, consumers need a shopping cart or basket when making their purchases of various items. In addition, the process of procuring the products is rather complicated as the shoppers must carry a basket every time in order to find the products they want and place them into the basket, followed by needing to deal with the calculation of the expenses. After completing their shopping list, the consumers must wait in a long line so that they can check out and pay the bill. In this paper, we describe a novel cost-effective method to overcome these issues by creating a smart trolley using an RFID Tag. In comparison with previous methods, our solution costs 10 times lesser than its predecessors and is environment friendly as well.

Keywords— Smart Shopping; Smart Trolley; RFID Transceiver.

I. INTRODUCTION

In this modern world, all people like to use products which are of high technology. People do not want to waste time and energy by using conventional systems. Rather they prefer advanced devices which are automated, smart, to finish their work soon. Smart trolleys are one such advanced device which is more flexible and an easy process to complete shopping without any delay. Customers in the store do not want to wait for a long time to pay their bill. In the smart trolley the bill can be paid simultaneously without waiting in the counter. Once the shopping is over payment is done through online or offline based on the customer. This flexibility is not provided by the existing trolley [1].

With the recent discussions of foreign direct investment (FDI) and due to westernization of cities in developing nations such

as India, there has been a rapid growth of shopping centers and shopping malls across countries. A breakthrough recently in this regard has also been the introduction of numerous web portals through which consumers can shop online, but these come along with their set of disadvantages. Safety and security are one of the most primary concerns for consumers whilst shopping online. Another major disadvantage is the inability of consumers to have a look at the product description or try out the product before purchase. Though money-back return policies do exist, they are quite a hassle for customers, and this has been intimidated by them through several surveys. Shopping centers or malls on the other hand form a great source of convenience to consumers, since all the products of varying choices are found under one roof.

II. EXISTING SYSTEM

The traditional trolley is different from a smart trolley. The traditional trolley is widely available in most supermarkets and is cheap. Comparatively, the smart trolley is sophisticated, expensive, and very limited in number because of its manufacturing cost.

As mentioned, a lot of research has been carried out to address the issues mentioned in the previous section. One of the first publications on this topic was in the year 1995. The authors discuss a methodology of creating a Personal Shopping Assistant (PSA) which can be connected to a trolley. The PSA's basic role is to communicate with the user and intimate the user the location of items based on voice recognition.

Nowadays people purchase a variety of items and put them in the trolley. After total purchasing one should approach counter for billing purpose. By using barcode, the cashier

prepares the bill which is a time-consuming process. This results in long queues at the billing counters.

III. PROPOSED SYSTEM

Smart trolley using RFID technology is a modern approach to enhance the overall shopping experience of the customers by automating the billing process and minimizing the time taken to purchase items in a supermarket or a retail store. This paper presents the design and implementation of a smart trolley system using RFID technology [2]. The proposed system uses RFID tags and readers to identify and track the products in the trolley, eliminating the need for manual scanning of barcodes. This smart trolley system reduces the time taken to checkout by automating the billing process and provides a more convenient shopping experience to the customers [3].

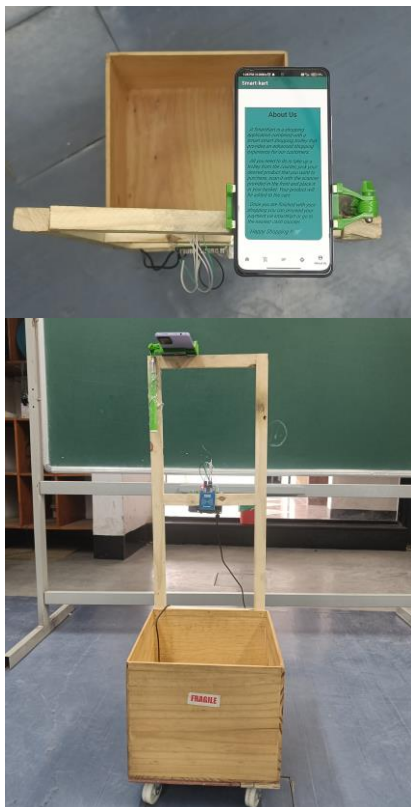


Fig 1: Proposed Trolley

IV. LITERATURE SURVEY

In this project, we discuss an innovative concept of RFID Based Smart Shopping and Billing System. The smart shopping trolley will help shorten the checkout lines thereby helping the customers at retail stores. The System consists of an RFID based trolley which communicates with the billing counter wirelessly using a RFID reader. The customers will be able to scan the items themselves and the LCD screen on the

shopping cart will keep updating the total. Every purchased products price is to be read by the RFID reader on the Smart cart through RFID tag on the product [7]. Every purchased products price is to be read by the RFID reader on the Smart cart through RFID tag on the product [8]. They transfer information from associate electronic tag, known as RFID tag or label, connected to associate object, through a reader for the purpose of identifying and tracking the object. A radio frequency identification reader (RFID reader) is a device accustomed gather data from an RFID tag that is employed to trace individual objects. Radio waves are accustomed transfer knowledge from the tag to a reader [12].

The smart trolley will consist of a RFID reader, LCD display and Raspberry-Pi microcontroller. When customer if want to buy any product is insert in the trolley. It will scan and read the product and display the cost and the name of the product in LCD. The total cost of all the purchased products will be added to the final bill [15].

In our project, earlier we made use of a camera module, with the help of which the images of the products were captured. Once the image has been captured, it is then run with our trained model. This trained model consists of all the datasets of all the products present in the store. Now this captured image when run with our trained model reveals the product details like name, cost, weight etc. So, instead of using barcode scanner or RFID scanner, our Smart trolley used image processing as its key tool, which helps in reducing the overall cost and most importantly it reduces the time to purchase a single product. It was built using a Camera, Barcode Scanner, Load Cell, Arduino Uno and Zigbee. [16] The prototype utilizes a unique color combination technique to map all products in a mall, without the use of any additional wireless sensors, with the least number of colors. Another advantage would be that it would reduce the time for the customers to locate their products as the system would locate it for them. [10]

V. SYSTEM DESIGN

The smart trolley system consists of RFID tags, RFID readers, a microcontroller, and a billing system. The RFID tags are attached to the products, and the RFID readers are mounted on the trolley. The RFID readers continuously read the tags and send the data to the microcontroller. The microcontroller processes the data and sends the billing information to the billing system. The billing system calculates the total cost of the items in the trolley and displays it to the customer.

The proposed system is designed to be user-friendly and easy to use. The customers can simply add items to the trolley and move around the store without worrying about the billing process. The RFID readers automatically read the tags, and the billing system calculates the total cost of the items in the trolley.

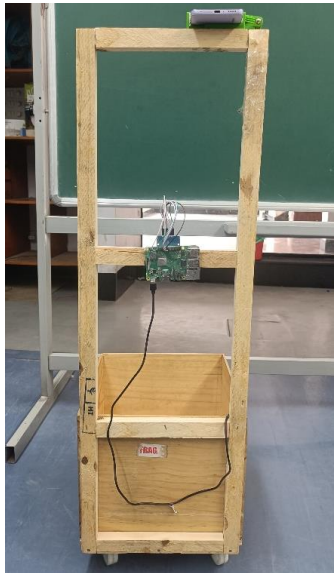


Fig 2: Trolley with Raspberry Pi and RFID

VI. TOOLS AND TECHNOLOGY

A. Hardware

Raspberry Pi 3 B+:

Raspberry Pi 3 Model B+ has an updated 64-bit quad core processor running at 1.4GHz with built-in metal heatsink, dual-band 2.4GHz and 5GHz wireless LAN, faster (300 mbps) Ethernet, and PoE capability via a separate PoE HAT. It has 1GB of RAM, faster dual-band 802.11 b/g/n/ac wireless LAN, Bluetooth 4.2, and significantly faster 300 Mbit/s ethernet. The Pi 3 B+ is to be provided with the same Pi as before, but now with Gigabit and PoE capable Ethernet, as well as better overheating protection for the 64-bit processor.



Fig 3: Raspberry Pi 3 B+

RFID Tags:

An antenna that delivers the data wirelessly to a reader and a microchip that contains identification information make up an RFID tag. At its most basic, the chip will provide a serialized identification, or number plate number, that, like many modern bar codes, uniquely identify that item. The fact that RFID tags can store more data than barcode tags is

a significant distinction. As a result, more information, such as the manufacturer, batch or lot number, weight, ownership, destination, and history (such as the temperature range to which an item has been exposed), can be encoded on the tag. Depending on the requirements of the application, RFID tags can store an infinite number of additional kinds of data. For identifying purposes, an RFID tag can be attached to individual objects, cases, or pallets as well as fixed assets like trailers, containers, totes, etc.



Fig 4: RFID Tags

RFID Scanner Module:

An RFID scanner is a device used to read and retrieve information from RFID (Radio Frequency Identification) tags or transponders. These tags contain a small microchip that stores information, and an antenna that allows the tag to communicate wirelessly with an RFID scanner.

RFID scanners work by emitting a radio signal that activates the RFID tag, causing it to transmit its stored information back to the scanner.

RFID scanners come in various types and sizes, from handheld devices to fixed installations. Some scanners can read multiple frequencies and protocols, making them compatible with a wide range of RFID tags. The data collected by the scanner can be processed and analyzed using software to gain valuable insights and improve operational efficiency.



Fig 5: RFID Scanner Module

LCD Display:

Here, we have used an LCD screen to display the application (Smartkart) that we have built. It is used to surf through the application and get a smart view of the cart.

B. Software

Android studio: Android Studio is the official integrated development environment for Google's Android operating system, built on JetBrains' IntelliJ IDEA software and designed specifically for Android development.

Java and Kotlin: Java and Kotlin is the preferred language for developing our Android app. Kotlin has been included as an alternative to the standard Java compiler.

Google Firebase: Firebase is a set of backend cloud computing services and application development platforms provided by Google. It hosts databases, services, authentication, and integration for a variety of application.

VII. IMPLEMENTATION USING CAMERA MODULE

Our Smart Shopping trolley previously included the concept of Machine Learning (ML). Basically, a small camera is mounted on the trolley which helps to capture an image of the product that the customer wishes to buy. Then this image is run in the database with the help of a microcontroller and then with the help of Machine Learning techniques the information about the product is gather and is displayed on an LCD display which is attached to the handle of the trolley [14].

The price of the product is determined with the help of separate barcodes assigned to each product. A barcode reader of an android smartphone will be included on the smart basket. In this way, the name and the cost of each item will be shown on the display of the LCD Tab when the customer scans and places the products in the basket. The total bill will be added up to give the total cost of all the purchases and stored to the memory of the microcontroller.

However, the model was trained in Roboflow platform which required the usage of Raspberry Pi 4, and the storage of Raspberry Pi 3b+ was not enough to contain the dataset. So, in order to make it simpler we switched the camera module with RFID.



Fig 6: Rpi Connected with Camera Module

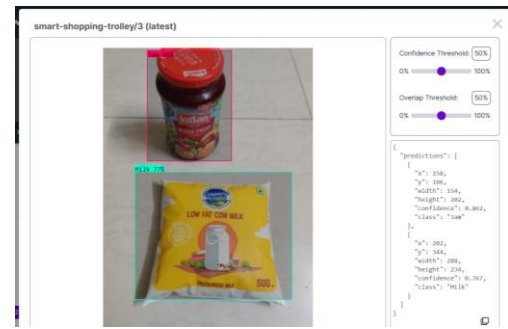


Fig 7: Model detecting the products



Fig 8: Camera detecting the Objects

VIII. IMPLEMENTATION USING RFID

The implementation of the smart trolley system involves the use of RFID tags and readers, a microcontroller, and a billing system. The RFID tags are attached to the products using adhesive stickers. The RFID readers are mounted on the trolley and are connected to the microcontroller. The microcontroller is programmed to process the data received from the RFID readers and send the billing information to the billing system. The customer's mobile application is initially connected to the trolley. The consumer can then begin scanning the products after that. The cart has an RFID reader attached to it, which reads the product tags. The application will show you a list of the products that have been scanned. Additionally, the entire bill and product cost will be displayed. The payment may be made via one of the available payments methods or it may also be made at the cash register.

The RFID tags and readers used in the system are passive RFID tags and readers. Passive RFID tags do not have an internal power source and rely on the power transmitted by the RFID readers. The RFID readers send a radio frequency signal to the RFID tags, which responds with the data stored in its memory. The RFID readers then receive the data and send it to the microcontroller for processing.

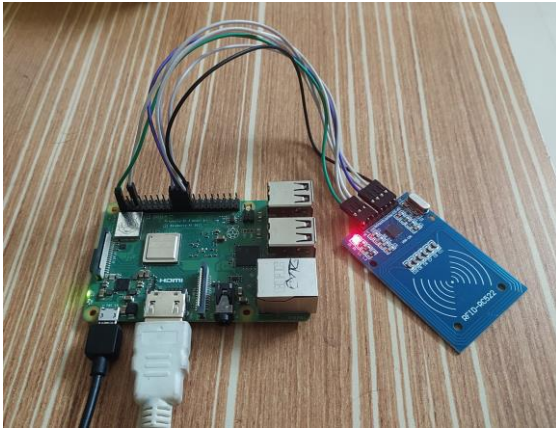


Fig 9: Raspberry Pi connected with RFID

IX. APPLICATION DESIGN

This is the front display of the application Smart-Kart in which several offers in the store is shown along with the categories of the products that are available in the store. This gives an idea to the customer about the latest products and offers that they can make use of. At the bottom of the front display, there are various tabs that are shown namely, Home, Cart, Notes, Map and About Us. Each tab has its own function.

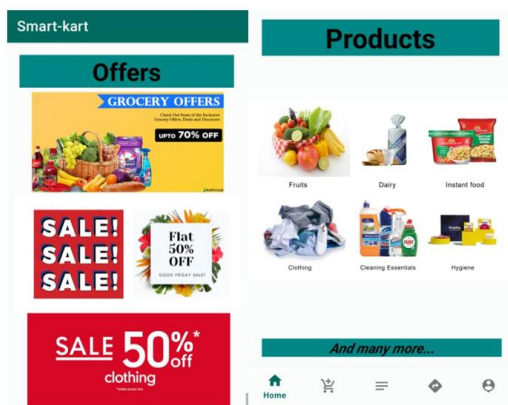


Fig 10: Home Page of Application

The second tab i.e., Cart, shows all the products which are added after being scanned by the RFID Scanner which is done by the customer. Once all the products are added into the cart, the customer will proceed by clicking next, where he/she can pay the required total amount by scanning the UPI QR Code with a displayed on the screen and pay, once done the bill will be paid. If the customer prefers to pay the bill via cash then they can go to the nearest cash counter to pay the bill.

In the Notes section the customer can add the required notes for e.g., if they want to make a list of the products they want to purchase.

A Static Map is provided if the customer wants to locate a specific alley in the store or if they are lost in the store then, he/she can get help from the static map provided in the application



Fig 11: Cart Section of the Application

X. SCOPE

The application of RFID is very precise and quick, even though it cannot completely replace the standardized identification innovation due to higher cost. The invention is currently being put into practice by companies like CVS, Tesco, Prada, Benetton, Wal-Shop, and Procter & Gamble, as well as manufacturers and merchants of consumer goods. RFID can also be used by others. Understanding innovation and many components to reduce potential problems is the foundation of success. It is time for the sector to start implementing new innovations, like RFID, in many applications, like manufacturing and storing things. The tags that are being used here can only be fastened to the products longitudinally, without any folds. Tags sent to the reader are only scanned when they are placed in the 3 to 4 cm range and they must be fastened to the products in a way that is visible to the reader in order for the reader to successfully detect tags. The tags used in this project are only capable of reading one side and are water and metal sensitive. The team can only now use products that are packaged with metal and water sensitive materials. However, the issue can be solved with the metal-resistant and waterproof tags now being researched. Additionally, the numerous objects the limitation of the RFID reader utilized here is that it occasionally skips the tags. According to the

requirements, a reader that can only identify 50 tags per second or less is used, keeping the system's installation costs in check.

XI. CONCLUSION

In this paper, we have designed and developed IOT based Smart Trolley System. The proposed smart trolley system using RFID technology is an efficient and convenient approach to automate the billing process in a retail store or a supermarket. The system eliminates the need for manual scanning of barcodes, reducing the time taken to checkout. The customers can simply add items to the trolley and move around the store without worrying about the billing process. The system is user-friendly and easy to use, providing a more convenient shopping experience to the customers. The system can be further enhanced by adding features such as an automated payment system, inventory management system, and customer tracking system.

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